
CYBER SECURITY INTRUSION DETECTION ANALYSIS

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Department: Computer Science and Engineering (CSE)

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Abstract

This project focuses on analyzing network traffic data to identify malicious activities, specifically focusing on Brute Force, Cross-Site Scripting (XSS), and SQL Injection attacks using the CICIDS2017 dataset. The methodology involves data cleaning, exploratory data analysis, SQL-based threat analytics, and the development of interactive dashboards using Python (Pandas, NumPy, Matplotlib, Seaborn) and Power BI. The primary objective is to enhance cybersecurity threat visibility, improve understanding of attack patterns, and provide efficient identification of suspicious entities through comprehensive reporting and visualization. Key findings indicate that Brute Force attacks constitute the majority of malicious traffic, while SQL Injection attacks are relatively rare.

Introduction

Cybersecurity is paramount in protecting computer systems, networks, and digital assets from unauthorized access, attacks, and data breaches. Intrusion Detection Systems (IDS) play a crucial role in monitoring network traffic and identifying suspicious behavior. This project aims to leverage data analytics techniques to extract meaningful insights from network traffic analysis. By focusing on the CICIDS2017 dataset, particularly the Thursday Web Attacks data, this study investigates various attack patterns to develop a robust intrusion detection and analysis framework. The project's objectives include analyzing network traffic data, detecting and classifying malicious activities, identifying high-risk source IPs, performing SQL-based threat analytics, creating interactive dashboards for visualization, and generating security recommendations.

Existing and Proposed System

Existing System

Traditional intrusion detection systems often rely on predefined signatures and manual monitoring techniques, which present several limitations. These include high false positive rates, difficulty in handling large-scale network traffic, limited visualization capabilities, complex interpretation of security logs, and a general lack of integrated analytical reporting. These shortcomings highlight the need for more advanced and automated solutions to effectively combat evolving cyber threats.

Proposed System

The proposed system offers a comprehensive cybersecurity threat analysis solution utilizing Python, SQL, and Power BI. This system aims to overcome the limitations of existing methods by providing enhanced threat visibility and a deeper understanding of attack patterns. Key features include robust data cleaning and preprocessing, detailed network traffic analysis, attack distribution analysis, source and destination IP analysis, protocol and port analysis, and risk score identification. The system will also incorporate interactive Power BI dashboards for intuitive reporting and visualization, coupled with actionable security recommendations. This approach facilitates efficient identification of suspicious entities and significantly improves the overall effectiveness of intrusion detection.

Software and Hardware Requirements

Software Requirements

- **Operating System:** Windows 10 / Windows 11
- **Programming Language:** Python 3.x
- **Libraries:** Pandas, NumPy, Matplotlib, Seaborn

- **Database:** MySQL
- **Visualization Tool:** Power BI Desktop
- **Development Environment:** Jupyter Notebook, VS Code

Hardware Requirements

- **Processor:** Intel Core i5 or higher
- **RAM:** Minimum 8 GB
- **Storage:** Minimum 20 GB free space
- **Display:** 1366 x 768 resolution or higher

Methodology

The methodology adopted for this project involves a structured approach to cybersecurity intrusion detection analysis:

1. **Dataset Collection:** The CICIDS2017 Thursday Web Attacks dataset was acquired as the primary data source.
2. **Data Cleaning:** This step involved removing duplicate records, handling missing values, addressing infinite values, converting timestamps, and standardizing column names to ensure data quality.
3. **Exploratory Data Analysis (EDA):** EDA was performed to understand the dataset's characteristics, including attack distribution analysis, traffic volume analysis, protocol analysis, and packet behavior analysis.
4. **SQL Analysis:** SQL queries were executed to identify specific attack distributions, analyze source and destination IP activity, evaluate protocol usage, and pinpoint high-risk IP addresses.
5. **Power BI Dashboard Development:** Interactive dashboards were developed using Power BI to visualize key insights, focusing on executive overview, traffic analysis, and attack analysis.

Algorithm

The cybersecurity threat analysis algorithm follows a systematic sequence of steps to process network traffic data and derive actionable intelligence:

1. **Load Dataset:** The network traffic dataset is loaded into the system.
2. **Perform Preprocessing:** Initial data cleaning and transformation are applied.
3. **Remove Invalid Records:** Records containing erroneous or irrelevant information are identified and removed.
4. **Handle Missing Values:** Strategies are implemented to address and impute missing data points.

5. **Conduct Exploratory Data Analysis:** Statistical and visual techniques are used to explore data patterns and anomalies.
6. **Store Cleaned Data in MySQL:** The processed and cleaned data is stored in a MySQL database for efficient querying.
7. **Execute SQL Threat Analytics:** SQL queries are run against the database to perform detailed threat analysis.
8. **Generate Visualizations:** Data visualizations are created to represent the findings effectively.
9. **Build Power BI Dashboards:** Interactive dashboards are constructed in Power BI for comprehensive data presentation.
10. **Interpret Findings:** The results from the analysis and visualizations are interpreted to generate threat intelligence insights.

Output: Threat intelligence insights and security recommendations.

Experiment Process

The experimental process was divided into several distinct phases to systematically build and evaluate the intrusion detection system:

- **Phase 1: Dataset Acquisition and Understanding:** Focused on obtaining the CICIDS2017 dataset and gaining a thorough understanding of its structure and content.
- **Phase 2: Data Cleaning using Python:** Involved writing Python scripts to clean and preprocess the raw dataset.
- **Phase 3: Feature Selection and Preprocessing:** Identified relevant features and further prepared the data for analysis.
- **Phase 4: Exploratory Data Analysis using Matplotlib and Seaborn:** Utilized Python libraries to perform in-depth exploratory data analysis and visualize initial patterns.
- **Phase 5: Importing Data into MySQL:** Loaded the cleaned and preprocessed data into a MySQL database.
- **Phase 6: Executing SQL Queries for Cybersecurity Analytics:** Developed and ran SQL queries to perform specific cybersecurity analytics.
- **Phase 7: Building Power BI Dashboards:** Designed and implemented interactive dashboards in Power BI to present the analytical results.
- **Phase 8: Generating Insights and Recommendations:** Derived actionable insights and formulated security recommendations based on the comprehensive analysis.

Result

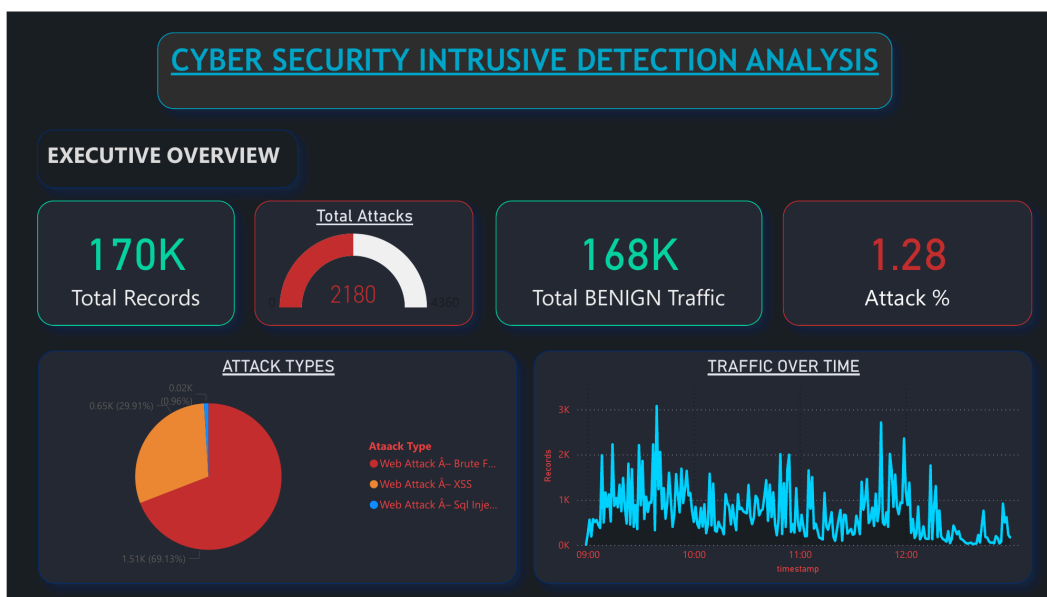
The analysis of the CICIDS2017 dataset yielded significant results regarding network intrusion detection. A total of over 170,000 records were analyzed, leading to the identification of more than 2,180 attacks, with benign traffic accounting for over 168,000 records. The overall attack percentage was approximately 1.28%. The distribution of attacks revealed that Brute Force attacks were the most prevalent, accounting for 69.13%, followed by XSS at 29.91%, and SQL Injection at 0.96%.

Major findings include the dominance of Brute Force attacks in malicious traffic and the identification of high-risk IP addresses. DNS (Port 53) and HTTPS (Port 443) traffic were identified as major communication channels, with TCP and UDP protocols being heavily utilized. Risk scores were calculated using TCP flag counts, aiding in the identification of suspicious source IPs.

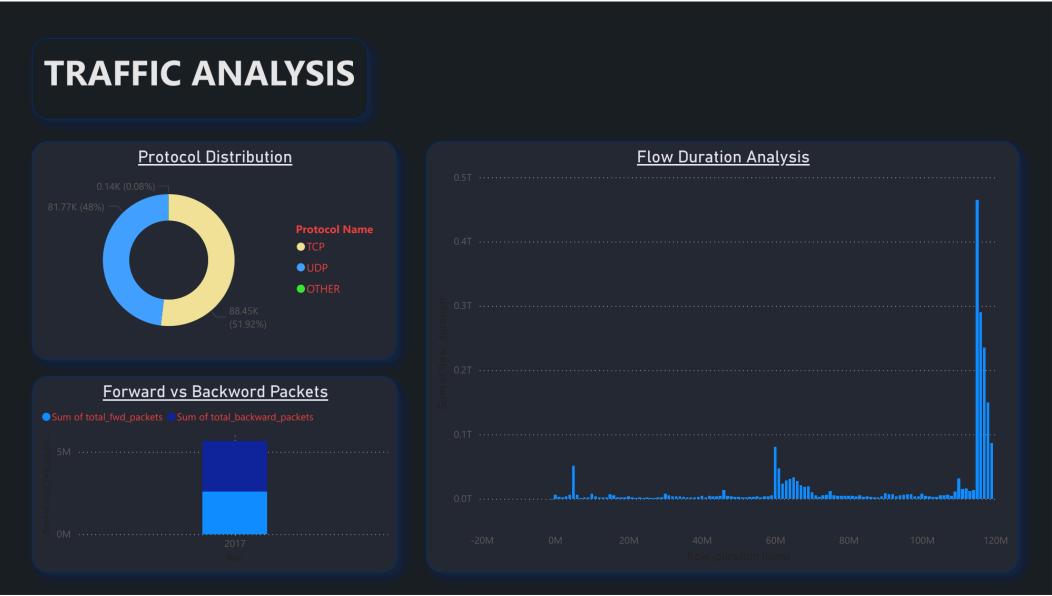
Output Screenshots

Power BI Dashboards

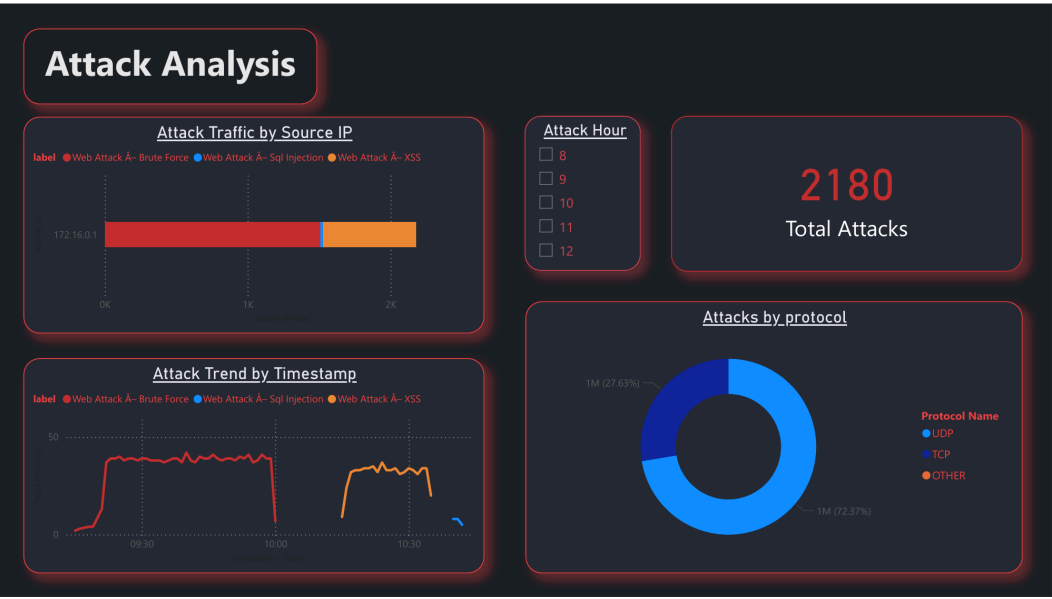
Screenshot 1: Power BI Executive Overview Dashboard



Screenshot 2: Traffic Analysis Dashboard



Screenshot 3: Attack Analysis Dashboard



SQL Threat Analytics

Screenshot 4: Attack Distribution SQL Output

```
mysql> select label as attack_type,count(*) as total_attacks from logs where label<>"BENIGN" group by label;
+-----+-----+
| attack_type | total_attacks |
+-----+-----+
| Web Attack A? Brute Force | 1507 |
| Web Attack A? XSS | 652 |
| Web Attack A? Sql Injection | 21 |
+-----+-----+
3 rows in set (0.12 sec)
```

Screenshot 5: High Risk Score Identification SQL Output

```
mysql> select source_ip, sum(syn_flag_count+ rst_flag_count + ack_flag_count) as risk_score from logs group by source_ip order by risk_score desc limit 10;
```

source_ip	risk_score
192.168.10.15	2814
192.168.10.58	2528
192.168.10.14	1484
192.168.10.17	1488
192.168.10.8	1248
192.168.10.9	1227
192.168.10.12	1223
192.168.10.5	1285
192.168.10.19	1139
192.168.10.16	990

10 rows in set (0.34 sec)

Screenshot 6: Protocol Analysis SQL Output

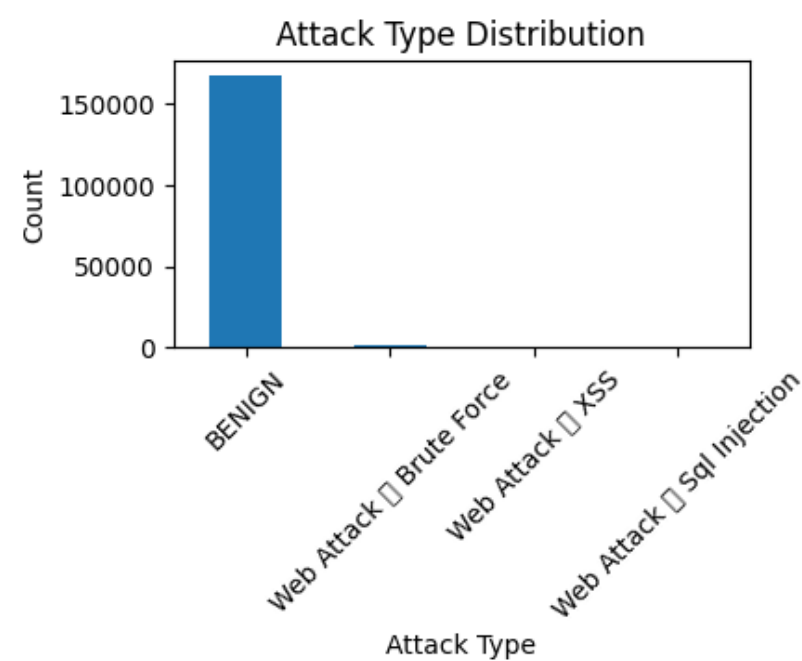
```
mysql> select protocol, count(*) as protocol_count from logs group by protocol order by protocol_count ;
```

protocol	protocol_count
0	84
17	56473
6	58605

3 rows in set (0.14 sec)

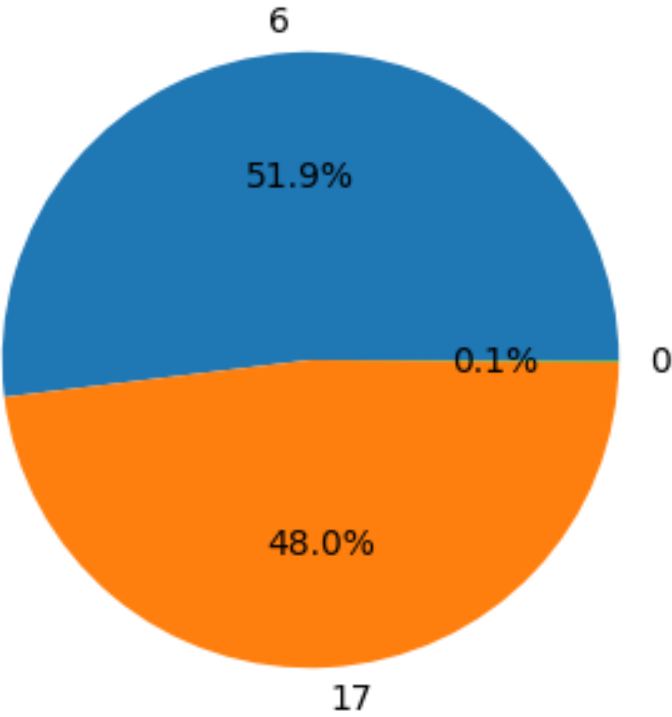
Python Visualizations

Screenshot 7: Attack Type Distribution

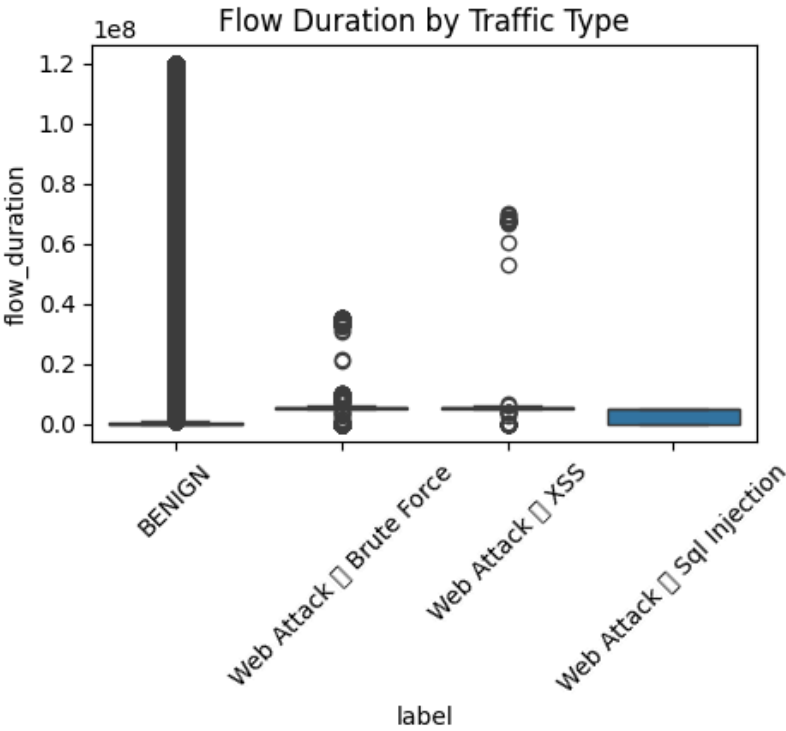


Screenshot 8: Protocol Distribution

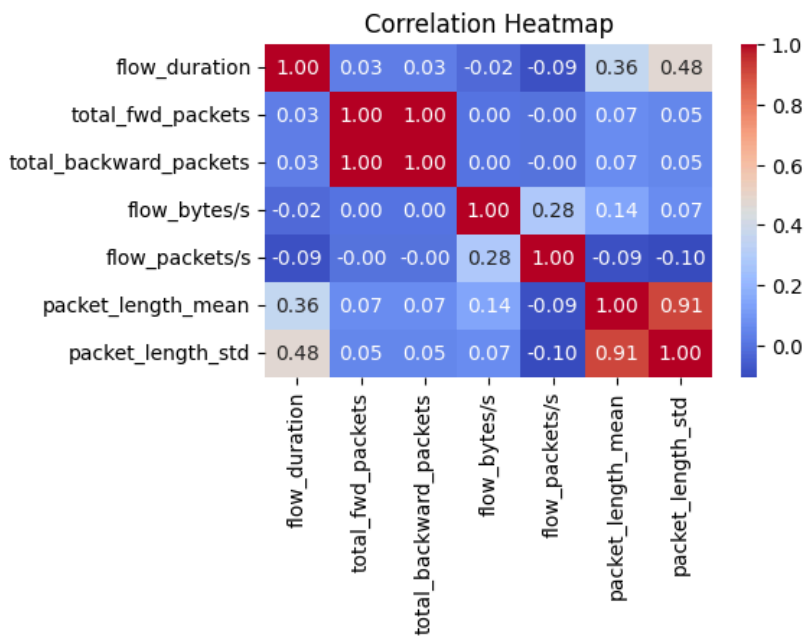
Protocol Distribution



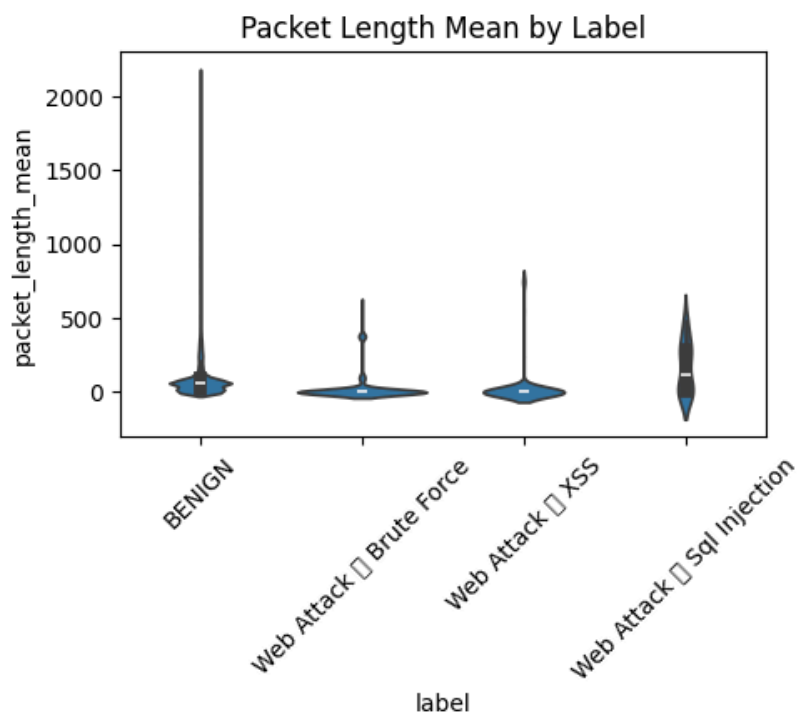
Screenshot 9: Flow Duration by Traffic Type



Screenshot 10: Correlation Heatmap



Screenshot 11: Packet Length Mean by Label



Conclusion

This project successfully demonstrated the application of data analytics techniques, utilizing Python, SQL, and Power BI, for effective cybersecurity intrusion detection. The comprehensive analysis of network traffic data from the CICIDS2017 dataset provided meaningful insights into various attack

patterns, including Brute Force, XSS, and SQL Injection. The developed framework offers enhanced threat visibility, improved understanding of attack behaviors, and efficient identification of suspicious entities through interactive dashboards and detailed reporting. The practical implementation of these data-driven analysis techniques underscores their value in modern security operations, enabling proactive measures and informed decision-making to strengthen network defenses.

References

1. Sharafaldin, I., Lashkari, A. H., & Ghorbani, A. A. (2018). Toward Generating a New Intrusion Detection Dataset and Intrusion Traffic Characterization. 4th International Conference on Information Systems Security and Privacy (ICISSP).
2. CICIDS2017 Dataset Documentation, Canadian Institute for Cybersecurity (CIC).
3. Python Documentation for Pandas, NumPy, Matplotlib, and Seaborn libraries.
4. Power BI Desktop Documentation, Microsoft.